UML Class Diagram Test Plan

UML Class: GameItem

UML Class Description: Base item metadata; parent of Key, Door and Potion.

Test date: _____ Tester: _____ Build / commit: _____

Fresh independent objects per case. Call the listed public methods in a separate test driver; application source stays unchanged. All results are pending execution.

TEST SCENARIO	TEST CASE	PRE-CONDITION	TEST STEPS	TEST DATA	EXPECTED RESULT	PASS / FAIL
Constructor / getters	UML-001	None.	Construct; call getName(), getSymbol(), getType().	GameItem("Coin","C","collectible")	Returns Coin, C, collectible respectively.	Not run
Exact metadata	UML-002	None.	Construct with empty strings; call all getters.	"", "", ""	All return empty strings unchanged; constructor performs no validation.	Not run

UML Class Diagram Test Plan

UML Class: Key

UML Class Description: Colored collectible extending GameItem.

Test date: _____ Tester: _____ Build / commit: _____

Fresh independent objects per case. Call the listed public methods in a separate test driver; application source stays unchanged. All results are pending execution.

TEST SCENARIO	TEST CASE	PRE-CONDITION	TEST STEPS	TEST DATA	EXPECTED RESULT	PASS / FAIL
Constructor / color	UML-003	None.	Construct and call getColor() and inherited getters.	Key("Red","R")	Color Red; name RedKey; symbol R; type collectible.	Not run
Blue key	UML-004	None.	Repeat with blue key.	Key("Blue","B")	Color Blue; name BlueKey; symbol B; type collectible.	Not run

UML Class Diagram Test Plan

UML Class: Door

UML Class Description: Colored passage extending GameItem; matches a key by exact color.

Test date: _____ Tester: _____ Build / commit: _____

Fresh independent objects per case. Call the listed public methods in a separate test driver; application source stays unchanged. All results are pending execution.

TEST SCENARIO	TEST CASE	PRE-CONDITION	TEST STEPS	TEST DATA	EXPECTED RESULT	PASS / FAIL
Constructor / color	UML-005	None.	Construct; call getColor() and inherited getters.	Door("Red","R")	Red, Red Door, R, door respectively.	Not run
Matching key	UML-006	Red door.	Call matchesKey with red, blue and lowercase red keys.	Key("Red","r"), Key("Blue","b"), Key("red","r")	Returns true, false, false; color matching is case-sensitive.	Not run
Null key	UML-007	Red door.	Call matchesKey(null).	null	Current behavior: NullPointerException; no null guard.	Not run

UML Class Diagram Test Plan

UML Class: Potion

UML Class Description: GameItem with an integer effect amount.

Test date: _____ Tester: _____ Build / commit: _____

Fresh independent objects per case. Call the listed public methods in a separate test driver; application source stays unchanged. All results are pending execution.

TEST SCENARIO	TEST CASE	PRE-CONDITION	TEST STEPS	TEST DATA	EXPECTED RESULT	PASS / FAIL
Constructor / effect	UML-008	None.	Construct; call getEffectAmount() and inherited getters.	Potion("Test","P","health",10)	Effect 10; name Test, symbol P, type health.	Not run
Effect boundaries	UML-009	None.	Construct with zero and negative amounts.	0, -5	getEffectAmount returns 0 and -5 unchanged; no validation.	Not run

UML Class Diagram Test Plan

UML Class: HealthPotion

UML Class Description: Potion subtype with fixed healing metadata.

Test date: _____ Tester: _____ Build / commit: _____

Fresh independent objects per case. Call the listed public methods in a separate test driver; application source stays unchanged. All results are pending execution.

TEST SCENARIO	TEST CASE	PRE-CONDITION	TEST STEPS	TEST DATA	EXPECTED RESULT	PASS / FAIL
Default constructor	UML-010	None.	Construct; call getEffectAmount() and inherited getters.	new HealthPotion()	Effect 25, name Health potion, symbol Unicode U+1F497 (heart), type health.	Not run

UML Class Diagram Test Plan

UML Class: SpeedPotion

UML Class Description: Potion subtype granting five additional moves.

Test date: _____ Tester: _____ Build / commit: _____

Fresh independent objects per case. Call the listed public methods in a separate test driver; application source stays unchanged. All results are pending execution.

TEST SCENARIO	TEST CASE	PRE-CONDITION	TEST STEPS	TEST DATA	EXPECTED RESULT	PASS / FAIL
Default constructor	UML-011	None.	Construct; call getEffectAmount() and inherited getters.	new SpeedPotion()	Effect 5, name speed potion, symbol Unicode U+26A1 (lightning), type speed.	Not run

UML Class Diagram Test Plan

UML Class: Player

UML Class Description: Position, health, collection flags and passed-door flags; parent of ComputerPlayer.

Test date: _____ Tester: _____ Build / commit: _____

Fresh independent objects per case. Call the listed public methods in a separate test driver; application source stays unchanged. All results are pending execution.

TEST SCENARIO	TEST CASE	PRE-CONDITION	TEST STEPS	TEST DATA	EXPECTED RESULT	PASS / FAIL
Constructor / position	UML-012	None.	Construct; call getRow(), getCol(), getHealth() and all flag getters.	Player(1,1)	Row 1, col 1, health 100; hasKey, hasRedKey, hasBlueKey, hasPassedRedDoor and hasPassedBlueDoor all false.	Not run
Move	UML-013	Player at (1,1).	Call move; inspect getRow and getCol.	move(3,4)	Position becomes (3,4). No board validation occurs in Player.	Not run
General key	UML-014	Fresh Player.	Call collectKey(); then hasKey().	No arguments	hasKey true; colored-key flags remain false.	Not run
Colored keys	UML-015	Fresh Player.	Call collectRedKey(), check flags; then collectBlueKey().	No arguments	hasRedKey true after first call; hasBlueKey true after second. General hasKey remains false.	Not run
Door flags	UML-016	Fresh Player.	Call passRedDoor(), then passBlueDoor(); inspect both getters after each.	No arguments	Only red flag true after first call; both passed-door flags true after second.	Not run
Damage / healing	UML-017	Fresh health 100.	takeDamage(40); heal(10); heal(); getHealth after each.	40, 10, default 25	Health 60, 70, 95 respectively; covers both heal overloads.	Not run
Health bounds	UML-018	Fresh health 100.	heal(25); takeDamage(150).	25, 150	Current behavior: health 125 then -25; no maximum cap or minimum clamp.	Not run

UML Class Diagram Test Plan

UML Class: Timer

UML Class Description: Move budget used to determine time-based loss.

Test date: _____ Tester: _____ Build / commit: _____

Fresh independent objects per case. Call the listed public methods in a separate test driver; application source stays unchanged. All results are pending execution.

TEST SCENARIO	TEST CASE	PRE-CONDITION	TEST STEPS	TEST DATA	EXPECTED RESULT	PASS / FAIL
Constructor / getter	UML-019	None.	Construct; call getMovesRemaining() and isDone().	Timer(25)	25 and false.	Not run
Decrease / done	UML-020	Timer(1).	decrease; inspect count and isDone; decrease again.	1 to 0 to -1	Counts 0 then -1; isDone true for both.	Not run
Add moves	UML-021	Timer(0).	addMoves(5); inspect count and isDone.	5	Count 5 and isDone false.	Not run
Nonpositive start	UML-022	None.	Construct with zero and negative values.	Timer(0), Timer(-2)	Exact count retained; isDone true in both cases.	Not run

UML Class Diagram Test Plan

UML Class: Dungeon

UML Class Description: 20 x 20 board, collision rules, tile effects, search and display.

Test date: _____ Tester: _____ Build / commit: _____

Fresh independent objects per case. Call the listed public methods in a separate test driver; application source stays unchanged. All results are pending execution.

TEST SCENARIO	TEST CASE	PRE-CONDITION	TEST STEPS	TEST DATA	EXPECTED RESULT	PASS / FAIL
Constructor / tiles	UML-023	None.	Construct; inspect getTile at known coordinates.	(1,1), (0,0), (17,18)	Player marker, wall, exit respectively.	Not run
Set / get tile	UML-024	Fresh dungeon.	setTile(1,2,"X"); getTile(1,2).	X	Returns X; neighboring tiles unchanged.	Not run
Move bounds	UML-025	Player(1,1).	Call isValidMove for out-of-range, wall and floor coordinates.	(-1,1),(20,1),(1,-1),(1, 20),(0,1),(1,2)	False for first five; true for final floor coordinate.	Not run
Door rules	UML-026	Fresh player without keys.	Check red/blue doors; collect red then blue key and repeat.	(17,15), (17,17)	Both false initially; only red true after red key; both true after both keys.	Not run
Board output	UML-027	Fresh dungeon.	Capture displayBoard().	No arguments	20 lines, each containing 20 tile symbols in board order.	Not run
Find tile	UML-028	Fresh dungeon.	Call findTile for red key, exit, floor and missing string.	Red key; exit; floor; missing	Returns [9,13], [17,18], [1,2], null respectively; first row-major match returned.	Not run
Key effects	UML-029	Fresh objects per color.	Move player to key; call checkTile.	Red (9,13); blue (17,16)	Matching key flag true; corresponding pickup message printed.	Not run
Potion effects	UML-030	Health 60; timer 25; independent fixtures.	Move onto health or speed tile; checkTile.	(15,9); (17,14)	Health becomes 85 or timer becomes 30, with corresponding message.	Not run
Trap / doors	UML-031	Independent fixtures.	Move onto trap or unlocked door; checkTile.	(2,4), (17,15), (17,17)	Trap subtracts 20 health; each door sets matching passed flag. checkTile itself does not enforce key ownership.	Not run

TEST SCENARIO	TEST CASE	PRE-CONDITION	TEST STEPS	TEST DATA	EXPECTED RESULT	PASS / FAIL
Neutral effects	UML-032	Health 100, timer 25.	checkTile at floor, monster, treasure and exit.	(1,2), (2,8), (17,9), (17,18)	No health/timer/flag change or effect message; win detection belongs to main.	Not run
Visible board	UML-033	Fresh player at (1,1).	Capture displayVisibleBoard under a 1-second timeout.	Player(1,1)	Intended target: nearby tiles visible, others hidden, with row breaks. Inspection identifies col=playerCol assignment and missing row braces; current code may loop indefinitely. Record failure if timeout occurs.	Not run
Invalid tile index	UML-034	Fresh dungeon.	Call getTile and setTile outside board.	getTile(-1,0); setTile(20,0,"X")	Current behavior: ArrayIndexOutOfBoundsException; unlike isValidMove, these methods do not guard indices.	Not run

UML Class Diagram Test Plan

UML Class: ComputerPlayer

UML Class Description: Player subtype with random, greedy and breadth-first path selection.

Test date: _____ Tester: _____ Build / commit: _____

Fresh independent objects per case. Call the listed public methods in a separate test driver; application source stays unchanged. All results are pending execution.

TEST SCENARIO	TEST CASE	PRE-CONDITION	TEST STEPS	TEST DATA	EXPECTED RESULT	PASS / FAIL
Constructor	UML-035	None.	Construct; inspect inherited getters.	ComputerPlayer(1,1, 1)	Position (1,1), health 100, flags false; subsequent move selection uses easy mode.	Not run
Random move	UML-036	Fresh dungeon and easy computer at (1,1).	Call chooseRandomMove repeatedly without moving player.	20 calls	Every result S or D, both legal; do not require both choices to appear in a finite sample.	Not run
Easy dispatcher	UML-037	Difficulty 1.	Call chooseMove(dungeon).	Start (1,1)	Returns S or D through random strategy.	Not run
Hard dispatcher	UML-038	Difficulty 2; fresh board.	Compare chooseMove and chooseStrategicMove without state changes.	Start (1,1)	Both return S; first best Manhattan-distance tie wins in W,S,A,D order.	Not run
Strategy targets	UML-039	Fresh fixture per stage.	Set flags and location, then chooseStrategicMove.	No key at (9,12); red key at (17,14); passed red door at (17,15); blue key at (17,16); both doors passed at (17,17)	Each returns D toward, respectively, red key, red door, blue key, blue door, exit. Set all prerequisite flags for each stage.	Not run
Path next step	UML-040	Fresh board; computer (1,1).	Call findPath(d,targetRow,targetCol).	(1,2); (2,1)	Returns D then S; does not move player or change board.	Not run
Path edge cases	UML-041	Fresh board; no keys.	Call findPath for start, wall, locked door and out-of-board target.	(1,1),(0,0),(17,15),(20,20)	Returns null for all four: already at target or no reachable target.	Not run

TEST SCENARIO	TEST CASE	PRE-CONDITION	TEST STEPS	TEST DATA	EXPECTED RESULT	PASS / FAIL
Trapped player	UML-042	Set all four adjacent tiles to walls.	Call chooseRandomMove; separately chooseStrategicMove.	Computer at (1,1)	Current behavior: nextInt(0) raises IllegalArgumentException; strategic fallback also fails. Record lack of no-legal-move handling.	Not run
Missing target	UML-043	Red key replaced by floor; player lacks red key.	Call chooseStrategicMove.	setTile(9,13,floor)	Current behavior: NullPointerException because findTile returns null; no missing-target guard.	Not run

UML Class Diagram Test Plan

UML Class: MyProgram

UML Class Description: Console entry point; orchestrates player choice, turns, board effects and outcomes.

Test date: _____ Tester: _____ Build / commit: _____

main(String[] args) is the only declared method. Console cases cross-reference the Interface Test Plan. Fixture branch tests must be kept outside application source. No execution results are claimed.

TEST SCENARIO	TEST CASE	PRE-CONDITION	TEST STEPS	TEST DATA	EXPECTED RESULT	PASS / FAIL
main / human	UML-044	Compiled classes; fresh process.	Invoke main with human selection and 25 legal alternating moves.	1; D,A repeated 12 times; D	Displays prompts and boards; ends normally with time-loss message. See UI movement cases for input validation.	Not run
main / computer	UML-045	Fresh process per difficulty.	Invoke main with computer selection and difficulty; bound runtime to 10 seconds.	2 then 1; 2 then 2	25 autonomous valid moves and game-over without further input; no crash. Losing does not imply a strategy test failure.	Not run
Turn order	UML-046	Fixture driver mirrors main loop.	Record original target tile; clear old position; move; decrease timer; checkTile; write player marker; evaluate outcome.	Speed tile, trap, exit fixtures	Speed net +4 moves; trap subtracts 20 health; original exit marker still triggers victory after tile overwritten.	Not run
Loss / win priority	UML-047	Fixture driver; final move.	Exercise exit with zero budget, then non-exit with zero budget and zero health.	Final move to exit; final move onto trap	Victory suppresses losses; otherwise time-loss message precedes health-loss message. Fixtures test branch logic, not a claimed full console playthrough.	Not run